OPENNESS OF VERSALITY FOR THE STACK OF COHERENT SHEAVES IN THE FLAT CASE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-coherent (canonical tag 08KB). cited at statement line 1547. Standing situation: S locally Noetherian, S = B, f flat.
- Definition artin-definition-openness-versality (canonical tag 07XQ). cited at statement line 1551. The property being established: openness of versality.
- Lemma artin-lemma-get-openness-obstruction-theory (canonical tag 0CYF). cited at proof lines 1606 and 1630. The openness criterion; its footnote reduces functoriality to fixed A.
- Lemma lemma-coherent-diagonal (canonical tag 08W6). cited at proof line 1608. Criterion condition (1): representable diagonal.
- Lemma lemma-coherent-RS-star (canonical tag 08LQ). cited at proof line 1609. Criterion condition (2): Rim-Schlessinger.
- Lemma lemma-coherent-limits (canonical tag 08KD). cited at proof line 1610. Criterion condition (3): limit preservation.
- Definition artin-definition-obstruction-theory (canonical tag 07YG). cited at proof line 1626. The obstruction theory axioms.
- Lemma defos-lemma-flat-ringed-topoi (canonical tag 08VW). cited at proof lines 1624 and 1634. Obstruction class and tangent identification T_x(M) = Ext^1.
- Lemma defos-lemma-functorial-ringed-topoi (canonical tag 0CYE). cited at proof line 1633. Functoriality of obstruction classes for fixed A.
- Lemma spaces-perfect-lemma-compute-ext (canonical tag 08JR). cited at proof line 1644. The perfect object K computing Ext^i for i <= 2.
- Lemma more-algebra-lemma-pseudo-coherent-tensor (canonical tag 0CYB). cited at proof line 1662. Product-commutation characterization of pseudo-coherent complexes carries the final step.

Source reference. Original definition 08KB: sources/stacks-chow/tags.txt, source lines 10881–10881. Cited at statement line 1547: the lemma is phrased in this situation; load-bearing.

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Source reference. Original definition 07XQ: sources/stacks-chow/tags.txt, source lines 10085–10085. Cited at statement line 1551: the property being established; load-bearing.

Source reference. Original lemma 0CYF: sources/stacks-chow/tags.txt, source lines 16240–16240. Cited at proof lines 1606 and 1630: the criterion applied and the footnote reduction for functoriality; load-bearing twice.

Source reference. Original lemma 08W6: sources/stacks-chow/tags.txt, source lines 11259–11259. Cited at proof line 1608: condition (1) of the criterion; load-bearing.

Source reference. Original lemma 08LQ: sources/stacks-chow/tags.txt, source lines 10928–10928. Cited at proof line 1609: condition (2) of the criterion; load-bearing.

Source reference. Original lemma 08KD: sources/stacks-chow/tags.txt, source lines 10881–10881. Cited at proof line 1610: condition (3) of the criterion; load-bearing.

Source reference. Original definition 07YG: sources/stacks-chow/tags.txt, source lines 10111–10111. Cited at proof line 1626: the obstruction theory axioms; load-bearing.

Source reference. Original lemma 08VW: sources/stacks-chow/tags.txt, source lines 11249–11249. Cited at proof lines 1624 and 1634: the obstruction class o(F, F x I, 1) and the tangent identification T_x(M) = Ext^1; load-bearing twice.

Source reference. Original lemma 0CYE: sources/stacks-chow/tags.txt, source lines 16239–16239. Cited at proof line 1633: condition (ii) functoriality for fixed A; load-bearing.

Source reference. Original lemma 08JR: sources/stacks-chow/tags.txt, source lines 10859–10859. Cited at proof line 1644: the perfect object K with functorial isomorphisms H^i(K x M) = Ext^i; load-bearing.

Source reference. Original lemma 0CYB: sources/stacks-chow/tags.txt, source lines 16236–16236. Cited at proof line 1662: the product-commutation step H^i(K x prod M_n) = prod H^i(K x M_n) is carried by this characterization (K is perfect, hence pseudo-coherent); load-bearing.

Lemma 0.2. *In Situation 08KB assume that S is a locally Noetherian scheme, $S = B$, and $f : X \to B$ is flat. Let $\mathcal{X} = \mathcal{C}oh_{X/B}$. Then we have openness of versality for $\mathcal{X}$ (see Artin's Axioms, Definition 07XQ).*

Second proof. This proof is based on Artin's Axioms, Lemma 0CYF. Conditions (1), (2), and (3) of that lemma correspond to Lemmas 08W6, 08LQ, and 08KD.

We have constructed an obstruction theory in the chapter on deformation theory. Namely, given an S-algebra A and an object x of $\mathcal{C}oh_{X/B}$ over $\mathrm{Spec}(A)$ given by $\mathcal{F}$ on X_A we set $\mathcal{O}_x(M) = \mathrm{Ext}^2_{X_A}(\mathcal{F}, \mathcal{F} \otimes_A M)$ and if $A' \to A$ is a surjection with kernel I, then as obstruction element we take the element

$$o_x(A') = o(\mathcal{F}, \mathcal{F} \otimes_A I, 1) \in \mathcal{O}_x(I) = \mathrm{Ext}^2_{X_A}(\mathcal{F}, \mathcal{F} \otimes_A I)$$

of Deformation Theory, Lemma 08VW. All properties of an obstruction theory as defined in Artin's Axioms, Definition 07YG follow from this lemma except for functoriality of obstruction classes as formulated in condition (ii) of the definition. But as stated in the footnote to assumption (4) of Artin's Axioms, Lemma 0CYF it suffices to check functoriality of obstruction classes for a fixed A which follows

from Deformation Theory, Lemma 0CYE. Deformation Theory, Lemma 08VW also tells us that $T_x(M) = \mathrm{Ext}^1_{X_A}(\mathcal{F}, \mathcal{F} \otimes_A M)$ for any A-module M.

To finish the proof it suffices to show that $T_x(\prod M_n) = \prod T_x(M_n)$ and $\mathcal{O}_x(\prod M_n) = \prod \mathcal{O}_x(M)$. Apply Derived Categories of Spaces, Lemma 08JR to the computation of the Ext groups $\mathrm{Ext}^i_{X_A}(\mathcal{F}, \mathcal{F} \otimes_A M)$ for $i \leq m$ with $m = 2$. We find a perfect object $K \in D(A)$ and functorial isomorphisms

$$H^i(K \otimes^{\mathbf{L}}_A M) \longrightarrow \mathrm{Ext}^i_{X_A}(\mathcal{F}, \mathcal{F} \otimes_A M)$$

for $i = 1, 2$. A straightforward argument shows that

$$H^i(K \otimes^{\mathbf{L}}_A \prod M_n) = \prod H^i(K \otimes^{\mathbf{L}}_A M_n)$$

whenever K is a pseudo-coherent object of $D(A)$. In fact, this property (for all i) characterizes pseudo-coherent complexes, see More on Algebra, Lemma 0CYB. $\square$